

CHAIN-OF-THOUGHT STATISTICAL AND QUALITATIVE ANALYSIS

Dark Patterns Project | Revised 21-variable codebook

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Revision summary. The binary codebook now contains 21 variables. Codes 20 and 21 capture invented consumer preferences and unsupported factual claims. One human annotator codes all 20 selected traces, for $20 \times 21 = 420$ decisions.

SECTION 1: Preliminary Statistical Analysis

Codebook-based logistic regression and recommendation-score modeling

All codebook variables are binary and are not necessarily mutually exclusive. Green rows identify variables added in this revision.

#	Category	Variable	Operational definition
1	Product attributes	mentions_product_attributes	Mentions functional attributes such as quality, price, specifications, durability, or suitability.
2	Social proof ID	mentions_activity_message	Mentions popularity, purchases, views, demand, or activity claims.
3	Social proof ID	mentions_rating	Mentions star ratings, numerical ratings, or rating volume.
4	Social proof ID	mentions_testimonial	Mentions reviews, testimonials, endorsements, or customer experiences.
5	Social proof ID	mentions_assurance	Mentions seals, badges, guarantees, certifications, or trust marks.
6	Social proof impact	uptake_activity_message	Treats an activity message as a reason to recommend or not recommend a product.
7	Social proof impact	uptake_rating	Treats a rating as a reason to recommend or not recommend a product.
8	Social proof impact	uptake_testimonial	Treats a testimonial as a reason to recommend or not recommend a product.
9	Social proof impact	uptake_assurance	Treats an assurance as a reason to recommend or not recommend a product.
10	Skepticism	questions_claim_validity	Questions whether a product-page claim is accurate or trustworthy.
11	Skepticism	recognizes_persuasion	Explicitly identifies marketing, social proof, manipulation, or persuasive intent.

SECTION 1: Binary Codebook (continued)

#	Category	Variable	Operational definition
12	Consumer advocacy	considers_price_value	Discusses price, affordability, or value for money.
13	Consumer advocacy	considers_suitability	Discusses whether the product fits the consumer's stated or unstated needs.
14	Consumer advocacy	considers_risk	Discusses financial, performance, safety, privacy, or other perceived risks associated with the product.
15	Decision influence	changes_original_decision	The reasoning trace reveals that the model went back on an original decision.
16	Other reasoning	evaluates_product_features	Brings up features mentioned in the product description and reasons with them.
17	Other reasoning	performs_comparison	Evaluates a feature or price against known product variants, explicitly named or otherwise, from model training data.
18	Other reasoning	explicit_character	Explicitly mentions phrases such as 'social proof' or 'dark pattern'.
19	Other reasoning	questions_branding	Mentions a lack of brand information.
20	Rationale validity	invents_consumer_preference	Attributes a need, constraint, or preference to the consumer that is not stated in the prompt or product context.
21	Rationale validity	makes_unsupported_claim	States a factual claim about the product, consumer, brand, or page that is not supported by the provided content.

Human annotation protocol

- Randomly select 20 eligible reasoning traces from the full pilot dataset.
- Eligibility requires `response_status == 'ok'` and `parse_status == 'ok'`.
- One human annotator codes all 20 traces across all 21 variables.
- The terminal interface keeps the reasoning visible, asks for 0 or 1, checkpoints every answer, and displays global progress over 420 decisions.
- Use the completed annotations as few-shot examples for the LLM-assisted coding protocol.
- Audit code coverage and manually review a stratified sample of model-coded rows before final inference.

Analysis sequence

- Create `coding_few_shots.csv` with the reasoning trace and all 21 human codes.
- Create `pilot_codebook.csv` with `response_id`, `reason`, and all 21 LLM-assisted codes.
- Run one mixed-effects logistic regression for each binary code.
- Regress recommendation score on the selected or full code set, with experimental controls.
- Use the results to inform, but not predetermine, Section 2 thematic discovery.

SECTION 1: Statistical Models

(1) Separate model for each binary code k

$$M_{ik} \sim \text{Bernoulli}(\pi_{ik})$$

$$\text{logit}(\pi_{ik}) = \alpha_{0k} + \alpha_{Fk} F_i + \sum_p \alpha_{pk} P_{ip} + \alpha_{Ck} C_i + b_{k,m[i]} + u_{k,j[i]}$$

F, the social-proof treatment indicators P, and C enter as fixed effects. Model m and product j enter as crossed random intercepts. Each code is estimated separately because the outcomes are binary and not mutually exclusive.

(2) Recommendation-score model

$$S_i = \beta_0 + \beta_F F_i + \sum_p \beta_p P_{ip} + \beta_C C_i + \sum_{k=1}^K \gamma_k M_{ik} + b_{m[i]} + u_{j[i]} + \epsilon_i$$

For the full specification, $K = 21$. A selected-code model may use a prespecified subset. Because code variables are measured after treatment exposure, their coefficients should be interpreted as associations unless stronger mediation assumptions are justified.

Preliminary score model

$$S_i = \beta_0 + \sum_{k=1}^{21} \gamma_k M_{ik} + \epsilon_i$$

Regularization and stability

Use elastic net to stabilize correlated predictors and select among overlapping codes. Cross-validation selects the mixing parameter and penalty. The regularized model is predictive and complements, rather than replaces, the inferential mixed-effects and robust OLS specifications.

$$\text{minimize: mean squared error} + \lambda [\rho \sum |\gamma_k| + (1-\rho)/2 \sum (\gamma_k^2)]$$

Interpretation guardrail

Experimental condition P is randomized, so $P \rightarrow \text{score}$ and $P \rightarrow \text{code}$ contrasts can support treatment-effect claims. Associations between post-treatment codes and recommendation score should not automatically be described as causal effects of reasoning.

SECTION 1: Required Scripts and Outputs

01_select_random_sample.py

- Randomly select 20 eligible rows from pilot1_calls.csv using globally unique response_id values.
- Retain complete source rows and write selected_20.csv with a reproducible seed.

02_annotate_few_shots.py

- Present each reasoning trace to one human annotator and request all 21 binary codes.
- Display global progress over $20 \times 21 = 420$ decisions and checkpoint after every decision.
- Write coding_few_shots.csv. Existing 19-code annotation files migrate with blank values for the two new codes.

03_llm_code_pilot.py

- Load OPENROUTER_API_KEY from the environment or local .env file.
- Use model openai/gpt-5.6-sol and compile the completed human annotations as few-shot examples.
- Require strict structured output containing response_id, reason, and all 21 binary codes.
- Process every eligible pilot row, retry transient failures, checkpoint each success, and resume without recoding completed response IDs.

04_run_regressions.do

- Merge pilot1_calls.csv and pilot_codebook.csv one-to-one by response_id.
- Fit 21 crossed-random-effects logistic models, the full mixed score model, robust OLS, and cross-validated elastic net.
- Write logs, saved estimates, and analysis data under stata_results/.

run_pipeline.py and final archive

- Wrap selection, one-human annotation, optional LLM coding, and optional Stata execution in a resumable terminal workflow.
- Prompt securely for an API key when needed and detect an available Stata installation.
- Create pipeline_outputs.zip containing available annotation, final codebook, and Stata outputs, plus OUTPUT_MANIFEST.txt.

SECTION 2: Unsupervised Thematic Discovery

Research questions

- What recurring and emerging themes appear in the reasoning traces?
- What decision-making processes and failure modes are not anticipated by the binary codebook?
- How heterogeneous are reasoning strategies across products, models, and experimental conditions?

Recommended pipeline

```
Reasoning traces -> sentence embeddings -> UMAP -> HDBSCAN -> BERTopic c-TF-IDF representations  
-> human interpretation
```

BERTopic is the organizing framework: its standard modular pipeline already uses embeddings, dimensionality reduction, clustering, and topic representation. HDBSCAN is therefore configured inside BERTopic rather than run as a redundant predecessor.

Method roles

Method	Role	Primary output
PCA	Linear dimensionality reduction and variance diagnostics.	Components and explained variance.
UMAP	Nonlinear reduction for clustering and visualization.	Low-dimensional document coordinates.
HDBSCAN	Density-based clustering with explicit noise detection.	Cluster labels, probabilities, and outliers.
BERTopic	Topic representation over embedding-derived clusters.	Topic words, sizes, documents, and labels.
Similarity metrics	Measure trace heterogeneity and possible mode collapse.	Within-group cosine-similarity distributions.

Design safeguards

- Treat Section 2 as exploratory; themes discovered on the full corpus are not confirmatory codebook hypotheses.
- Inspect clusters within, or adjust for, product, model, condition, and trace length so surface content does not masquerade as reasoning structure.
- Have a human name and interpret clusters using representative and boundary cases, including HDBSCAN outliers.
- Validate any newly discovered code on a held-out or separately annotated sample before confirmatory use.

Existing literature on CoT analysis

Review the DP notebook and cited literature before finalizing model choices, embedding models, hyperparameters, and interpretation language. Visible reasoning traces support analysis of expressed rationales; they should not automatically be treated as direct observations of private model computation.